

CS & IT ENGINEERING



Computer Network - 1

Transport Layer

Lecture No. – 06

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Recap of Previous Lecture



Topic

TCP Data Transfer





Topics to be Covered



Topic

TCP Data Transfer



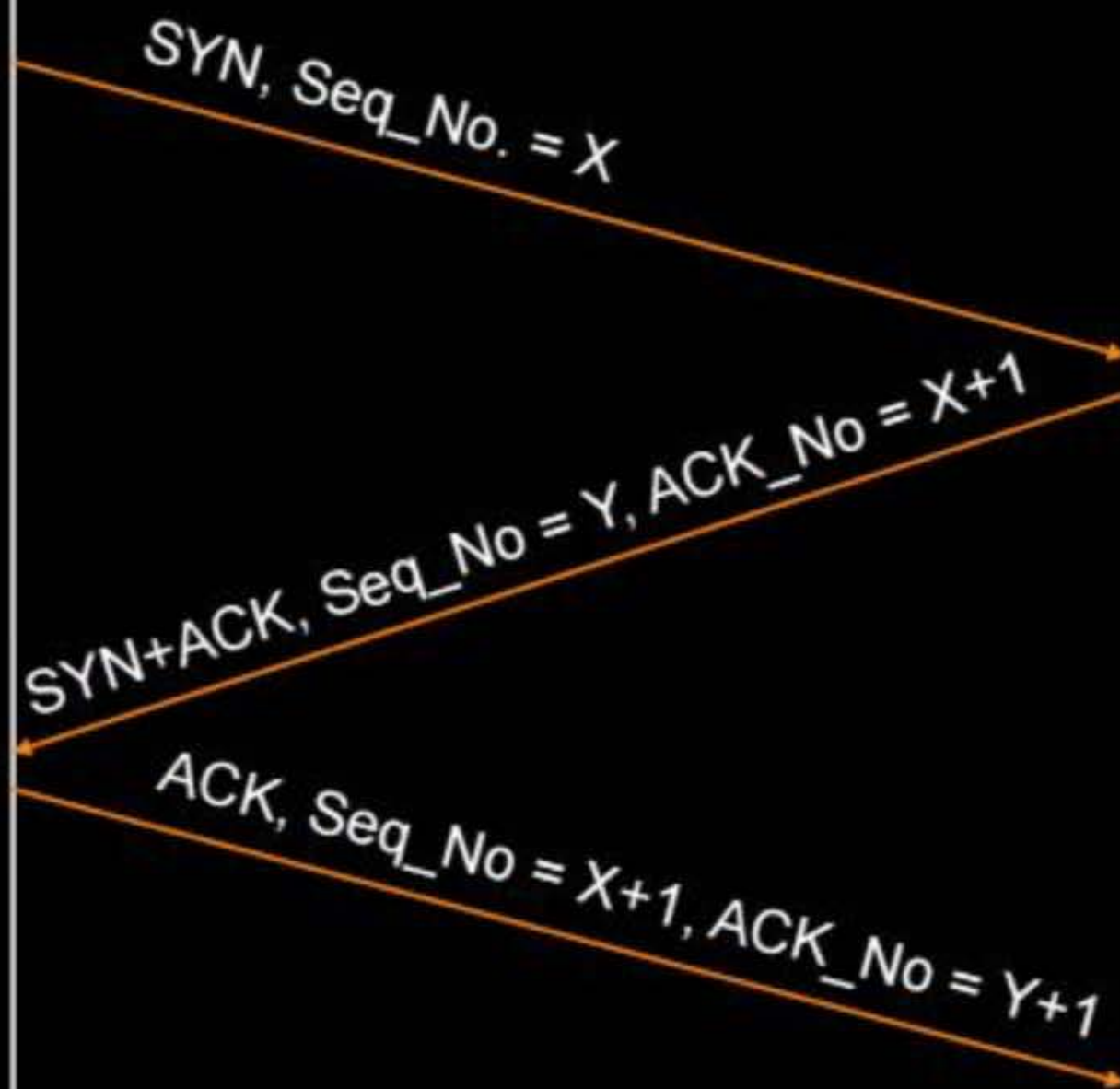
Client_ISN = X

(X=299)

TCP Client

TCP Server

Server_ISN = Y





Topic : TCP

Client_ISN = 299

MSS = 100 bytes



TCP Client

TCP Server

Server_ISN = Y

Seq.=300 [300-399], ACK = Y+1
Seq.=400 [400-499], ACK = Y+1
Seq.=500 [500-599], ACK = Y+1

Seq.=Y+1, ACK = 400
Seq.=Y+1, ACK = 500
Seq.=Y+1, ACK = 600



Topic : Sequence Number



=> TCP is byte-stream protocol (Stream-oriented protocol)

=> Sock_STREAM

=> Every byte in 'service data unit' is identified uniquely
[with 32-bit sequence number]

=> Unique sequence number is assigned to each byte

Sequence Number in TCP Segment ← First Byte Sequence Number of Payload

TCP → Every byte is special.

[MCQ]

IIT-D

[GATE-IT-2004] [1 Mark]



#Q. In TCP, a unique sequence number is assigned to each :

- ☒ A byte
- ☐ B word
- ☐ C segment
- ☐ D message

Ans: A



Topic : TCP

Client_ISN = 299

MSS = 100 bytes

TCP Client

TCP Server



Server_ISN = Y

Seq.=300 [300-399], ACK = Y+1
Seq.=400 [400-499], ACK = Y+1
Seq.=500 [500-599], ACK = Y+1

(Seq.=Y+1) ACK = 400
(Seq.=Y+1) ACK = 500
(Seq.=Y+1) ACK = 600

(300-399) ✓

(400-499) ✓

(500-599) ✓

← 300

← 400

← 500

← 600

TCP Server Process

P_2

SOCK_STREAM
(TCP Connection
Socket 1)

* $\text{LastByteWritten} = P_2\text{'s ISN} = Y$

✓ { $\text{LastByteSent} = P_2\text{'s ISN} = Y$
 $\text{LastByteAcked} = P_2\text{'s ISN} = Y$

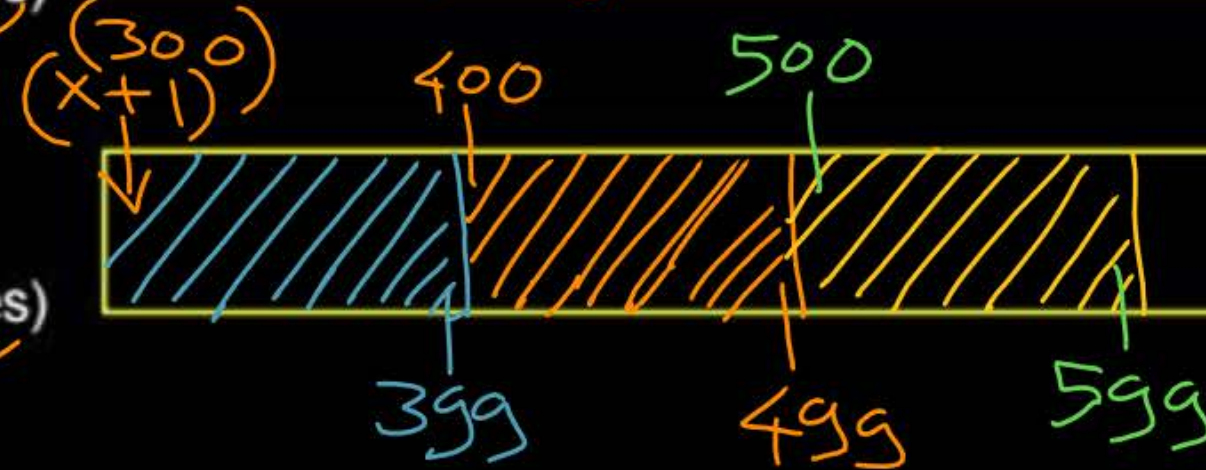
Send Buffer

(MaxSendBuffer = 1200 Bytes)

Empty

Recv Buffer

(MaxRecvBuffer = 1200 Bytes)



* $\text{LastByteRead} = P_1\text{'s ISN} = x = 299$

✓ { $\text{NextByteExpected} = P_1\text{'s ISN} + 1 = \cancel{300} = \cancel{400} = \cancel{500} = 600$
 $\text{LastByteRcvd} = P_1\text{'s ISN} = x = \cancel{299} = \cancel{399} = \cancel{499} = 599$



Topic : TCP



=> The TCP protocol receive a segment (contains K bytes in its payload) from process P₁, for process P₂

→ Store payload bytes in process P₂ 'Recv Buffer'

→ Update variables according to case and send ACK

=> Send Acknowledgement to sender TCP

Sequence Number = (LastByteSent + 1;)

ACK Number = NextByteExpected;



Topic : TCP

$(K=100)$



CASE I :

if (Sequence Number = NextByteExpected) and
(LastByteRcvd + 1 = NextByteExpected)

- Keep payload bytes in process P₂ 'Recv Buffer'
- NextByteExpected = NextByteExpected + K;
- LastByteRcvd = LastByteRcvd + K;



Topic : TCP



CASE II :

if Sequence Number < NextByteExpected

- ignore received TCP segment (duplicate segment)
- send Ack;

CASE III :

if (Sequence Number > NextByteExpected)

- Keep payload bytes in process P₂ 'Recv Buffer'
after [Sequence Number - NextByteExpected] empty bytes
- LastByteRcvd = Sequence Number + K - 1;

K = 100



Topic : TCP

Missing Byte = ~~200~~
100 K = 100



CASE IV:

500

500

if (Sequence Number = NextByteExpected) and
(LastByteRcvd > NextByteExpected) and
(Missing bytes > K)

799 > 500

→ NextByteExpected = NextByteExpected + K;

CASE V:

if (Sequence Number = NextByteExpected) and
(LastByteRcvd > NextByteExpected) and
(Missing bytes = K)

→ NextByteExpected = LastByteRcvd + 1;

TCP Server Process

P_2

SOCK_STREAM
(TCP Connection
Socket 1)

→ LastByteWritten = P_2 's ISN = Y

{ LastByteSent = P_2 's ISN = Y

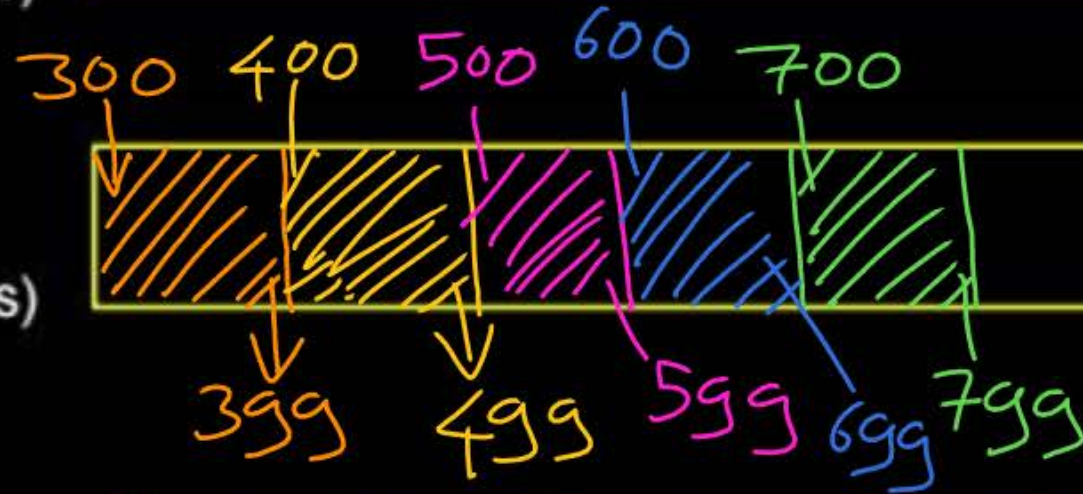
{ LastByteAcked = P_2 's ISN = Y

Send Buffer

(MaxSendBuffer = 1200 Bytes)

Recv Buffer

(MaxRecvBuffer = 1200 Bytes)



→ LastByteRead = P_1 's ISN = ~~x = 299~~ | 499

{ NextByteExpected = P_1 's ISN + 1 = ~~300~~ | ~~400~~ | ~~500~~ | ~~600~~ | 800

{ LastByteRcvd = P_1 's ISN = ~~299~~ | ~~399~~ | ~~499~~ | 799



Topic : TCP

P₁

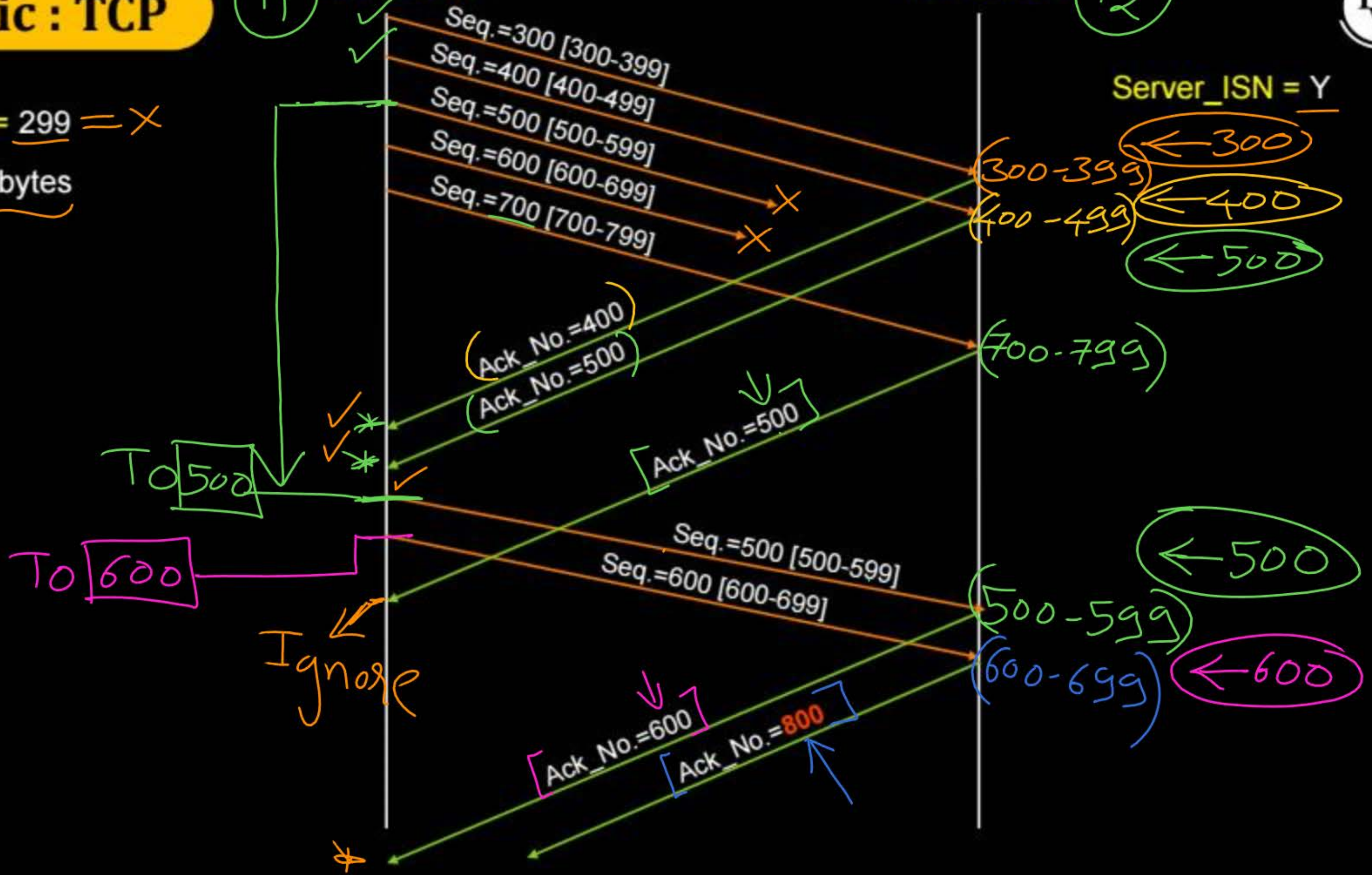
TCP Client

TCP Server

P₂



Client_ISN = 299 = X
MSS = 100 bytes





Topic : TCP



→ The process P_2 read N bytes from 'Recv Buffer'

→ $\text{LastByteRead} = \text{LastByteRead} + N$;

→ Remove N bytes from P_2 'Recv Buffer' from begin



Topic : Acknowledgment Number



=> if ACK flag is on
then TCP segment carry acknowledgment number
else :
Acknowledgment number is garbage value

In
Base header
(32-bit ACK No.)

=> TCP uses cumulative acknowledgment
[ACK Number = NextByteExpected ;]

(Combine ACK)





Topic : Acknowledgment Number



=> In TCP, provision for 'selective acknowledgment'
[TCP keeps 'selective acknowledgment(s)' in optional header]

=> TCP allows for piggy-backing ✓
[sending data and acknowledgments in the same segment]

$$[tx_A = tx]$$

$$[tx_A < tx]$$



Topic : TCP

Client_ISN = 299

MSS = 100 bytes

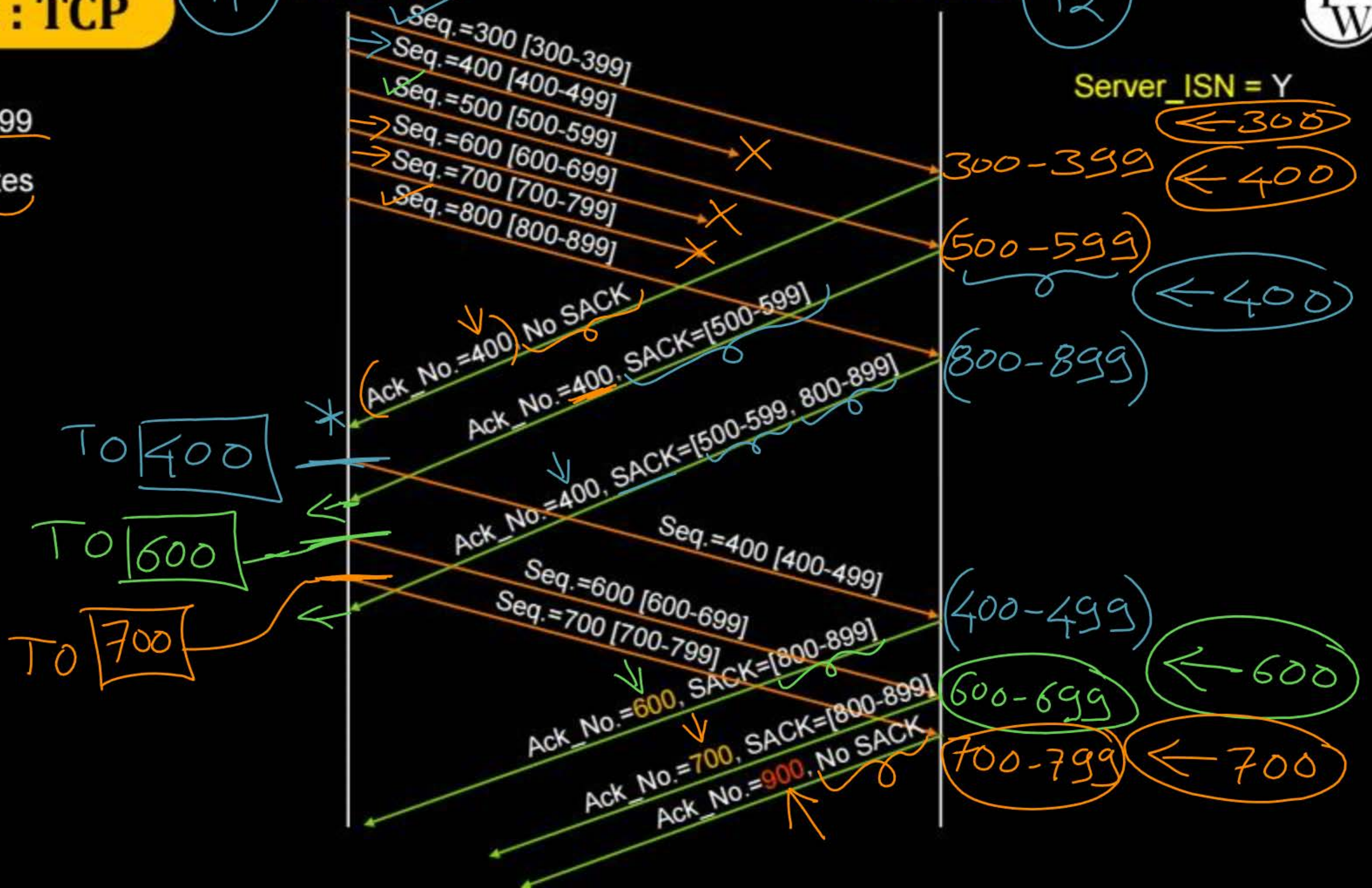
P₁ TCP Client

TCP Server

P₂



Server_ISN = Y



[MCQ]

[GATE-2015] [1 Mark]



#Q. Consider the following statements.

I. TCP connections are full duplex. TRUE

II. TCP has no option for selective acknowledgment FALSE

III. TCP connections are message streams. FALSE (Byte stream)

IIT-K

☒ A Only I is correct

☐ B Only I and II are correct

☐ C Only II and III are correct

☐ D All of I, II and III are correct

Ans: A

[MCQ]

IIT-K, H.W.

[GATE-IT-2007] [1 Mark]



#Q. Consider a TCP connection in a state where there are no outstanding ACKs. The sender sends two segments back to back. The sequence numbers of the first and second segments are 230 and 290 respectively. The first segment was lost, but the second segment was received correctly by the receiver. Let X be the amount of data carried in the first segment (in bytes), and Y be the ACK number sent by the receiver. The values of X and Y (in that order) are :

- A 60 and 290
- B 230 and 291
- C 60 and 231
- D 60 and 230



Topic : TCP

P1

TCP Client

TCP Server



Client_ISN = 299 = X

MSS = 100 bytes

→ 1600

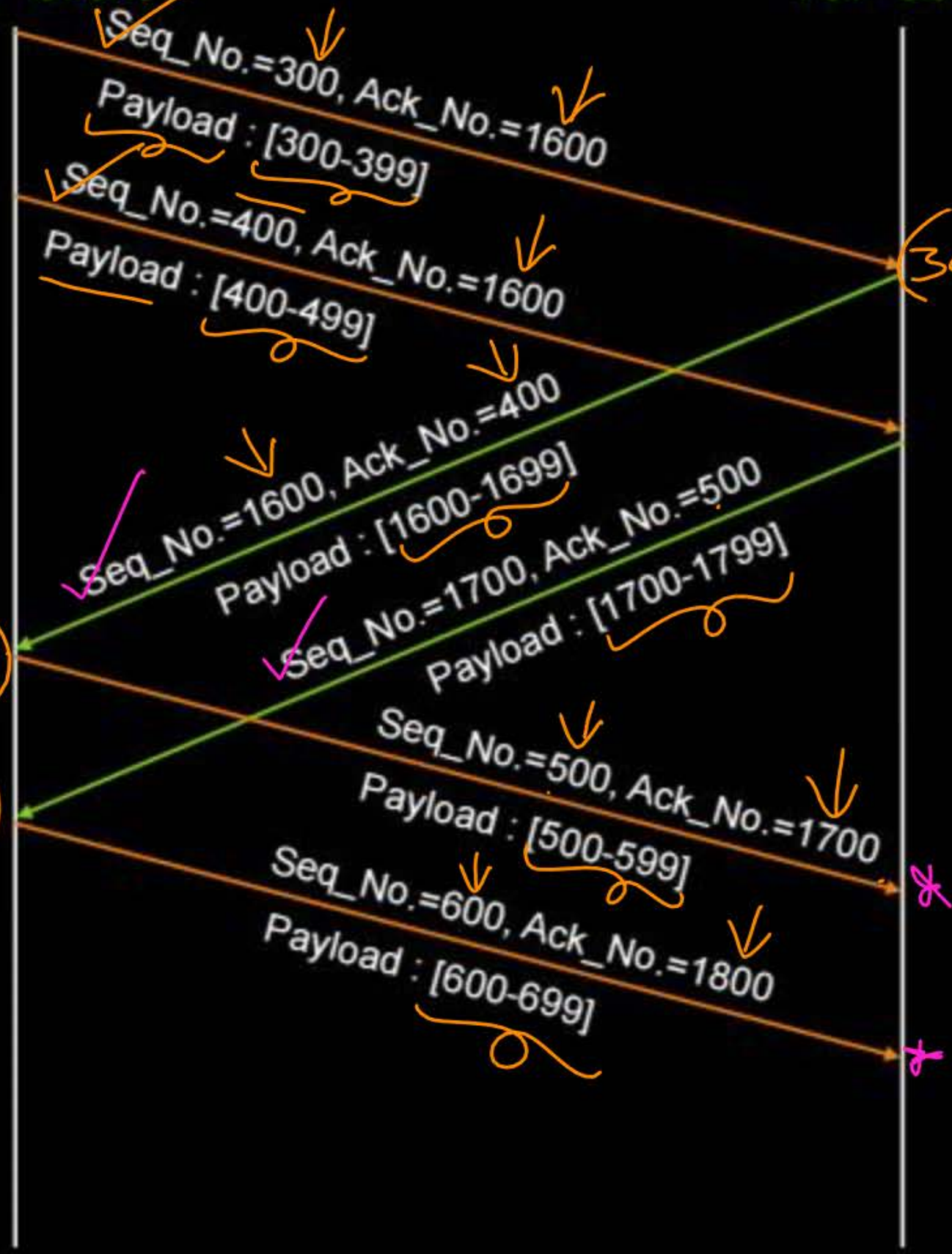
→ 1700

→ 1800

(1600-1699)
(1700-1799)

Server_ISN = 1599

= Y
← 300
← 400



[MCQ]

(IIT-R)

[GATE-2025] [1 Mark]



#Q. Consider the following statements :

(i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address.

(ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S.

Which ONE of the following is CORRECT?

- A Both (i) and (ii) are TRUE
- B (i) is TRUE and (ii) is FALSE
- C (i) is FALSE and (ii) is TRUE
- D Both (i) and (ii) are FALSE

(ii → T/F)
(ii → False)



2 mins Summary



Topic

TCP Data Transfer →



THANK - YOU

